

Jorge Ibáñez Puertas

Data Scientist | Machine Learning | Python & SQL

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Mathematician (BSc) with an MSc in Artificial Intelligence and a year of professional experience building data models, SQL pipelines and BI reporting used in production at client companies. I build things end to end, from raw data through modelling to a decision someone acts on. Academic distinction in pure mathematics, published applied optimisation research, and a first-principles approach to unfamiliar problems. Skilled in Python, SQL, TensorFlow, gradient boosting and mathematical optimisation.

AREAS OF EXPERTISE

Python • SQL • Pandas • NumPy • TensorFlow • Keras • Gradient Boosting • Time-Series Forecasting • Data Pipelines • Data Modelling & Warehousing • Statistical Inference • Mathematical Optimisation • AMPL • R • MATLAB • Git • LaTeX

PROJECTS

Code, write-ups and full publication list: jmompou.github.io/cv

Electricity Demand Forecasting from Low-Cost Signals | MSc Thesis, VIU Feb 2026 - Aug 2026

- Built an end-to-end forecasting pipeline in Python predicting electricity demand from inexpensive open-source proxy signals rather than costly sensor data, training a LightGBM gradient boosting model on 88,111 rows spanning 10 years of hourly observations.
- Engineered 18 calendar, lag, and weather-proxy features and ran walk-forward validation across 7 forecast horizons (1 to 168 hours), cutting day-ahead mean absolute error to 1,733 MW from 4,174 MW for the seasonal-naive baseline, and also beating the grid operator's own official day-ahead forecast (1,733 MW vs. 1,893 MW).
- Benchmarked the model against the seasonal-naive and official TSO forecasts and documented the accuracy-versus-data-cost trade-off, showing low-cost open signals recover most of the accuracy of proprietary telemetry at a fraction of the input cost.

Irrigation Coverage Optimisation | University of Murcia, with a working farm in Albacete 2023 - Present

- Formulated and solved a mixed-integer nonlinear (MINLP) model in AMPL/Gurobi (145.9 ha parcel, 261 variables, 462 constraints) to find the cost-minimal number, placement and size of center-pivot irrigation systems for a working farm, from field data gathered on site with the farm owner (Explotación Agraria Puertas SL, Albacete).
- Co-authored the resulting paper, published by the University of Murcia: Clemente Pérez, A., Ibáñez Puertas, J. & Colchero Truniger, N. (2023), *Ampliación del área de riego en una finca*.
- Independently extended the project: reimplemented the model natively in Python (Pyomo, no AMPL dependency), added automatic farm-boundary extraction from satellite imagery, and am now reframing pivot placement as a reinforcement-learning problem (PyTorch PPO) that places pivots on a new farm shape in a single forward pass instead of re-solving the MINLP from scratch; released as open source on [GitHub](https://github.com).

Inventory Models: Analysis and Applications | BSc Thesis, UNED 2025

- Studied and mathematically modeled deterministic (EOQ and variants: lead times, quantity discounts, multi-item, storage constraints) and probabilistic inventory systems (continuous and periodic review, multi-period, safety stock), including perturbation methods for inspection errors and returns, plus a supporting literature review. Grade: 9.6/10.

Almazen | Inventory Management Web App [YEAR]

- Built a browser-based tool that implements 5 inventory policies (EOQ/Wilson, EOQ + safety stock, periodic review, single-period/newsvendor, min-max s-S) to generate reorder-point and order-quantity suggestions from real stock data, with movement tracking and demand forecasting (SARIMA-style).

An Unusual Path | Independent research paper 2014

- Wrote an independent mathematics research paper that won the VIII Prize for Secondary School Students awarded by the Department of Mathematics, Universidad Autónoma de Madrid.

PROFESSIONAL EXPERIENCE

Unidata Murcia, Spain

Data Analyst and Developer June 2025 - November 2025

- Designed data models and wrote SQL queries and stored procedures to extract, transform and load data into the reporting layer behind real-time dashboards.
- Worked alongside software engineers on business applications running in production.

NTT Data Murcia, Spain

Data Analyst Intern June 2022 - September 2022

- Processed large volumes of data in Python and SQL and built Tableau dashboards for client reporting over a 3-month internship.

University of Murcia, Faculty of Mathematics Murcia, Spain

Teaching and Research Assistant (internship) December 2021 - June 2022

- Supported teaching and research in Multivariable Calculus and Vector Analysis, preparing course material for students.

EDUCATION

Universidad Internacional de Valencia (VIU) | MSc in Artificial Intelligence Oct 2025 - Aug 2026

- Neural networks, computer vision, deep learning, machine learning and data analysis. Grade: 7.75/10.

UNED, Spain | BSc in Mathematics 2018 - 2025

- *Matrícula de Honor* (highest distinction) in Discrete Mathematics and Graph Theory.
- Completed alongside full-time work; exchange year at Universidad Autónoma de Madrid, 2022-23.

Delft University of Technology, Netherlands | Aerospace Engineering, first year 2017 - 2018

- Admitted on academic merit following the International Baccalaureate Diploma with Distinction (2017); transferred to Mathematics to follow my main interest.

COURSES AND LANGUAGES

- **Courses:** Data Analysis and Visualisation: Practical Statistics with R and Artificial Intelligence (UNED, September 2025).
- **Languages:** Spanish (native); English C2, Cambridge Proficiency (CPE), 2026, Use of English 230/230.